

PERFORMANCE EVALUATION AND DIMENSIONING OF NETWORKS AND SYSTEMS

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Lab 1 Report [EN]

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1 Introduction

This report documents the work developed for Laboratory 1 of the Performance Evaluation and Dimensioning of Networks and Systems (DDRS) course, covering the fourteen exercises proposed in the laboratory guide. The exercises progress from the fundamentals of random number generation to the construction and analysis of discrete-event simulators, and finally to statistical techniques for validating and interpreting simulation output.

The report follows the same structure as the laboratory guide, organized into six parts. Part A covers the properties of R's pseudo-random number generator. Part B addresses the generation of random variables and stochastic processes – Bernoulli and exponential random variables, and Poisson arrival processes – from a uniform random number generator. Part C develops discrete-event simulators for the M/M/1 and M/M/2 queuing systems and compares their output against the corresponding theoretical performance metrics. Part D estimates a probability through Monte Carlo simulation. Part E applies statistical tools – histograms, confidence intervals, and an assessment of confidence-interval coverage – to both real network measurements (packet captures and RTT measurements) and synthetic data. Part F examines practical aspects of simulation output analysis: run-to-run variability, the effect of initial conditions, and the replication/deletion procedure for determining an adequate number of replications.

For each exercise, the report states the objective, describes the approach and the corresponding R script, and presents the results together with a discussion of what can be concluded from them. All scripts were developed in R and executed in RStudio; network traffic was captured with Wireshark.

2 A. Random number generators

2.1 Exercise 1

Objective: Generate 10000 random numbers uniformly distributed in $[0, 1]$ using `runif()` and plot their histogram with `hist()`, to check informally whether the generator behaves as expected.

Approach: A sample of 10000 values was generated from the uniform distribution on $[0, 1]$ and summarized with a histogram.

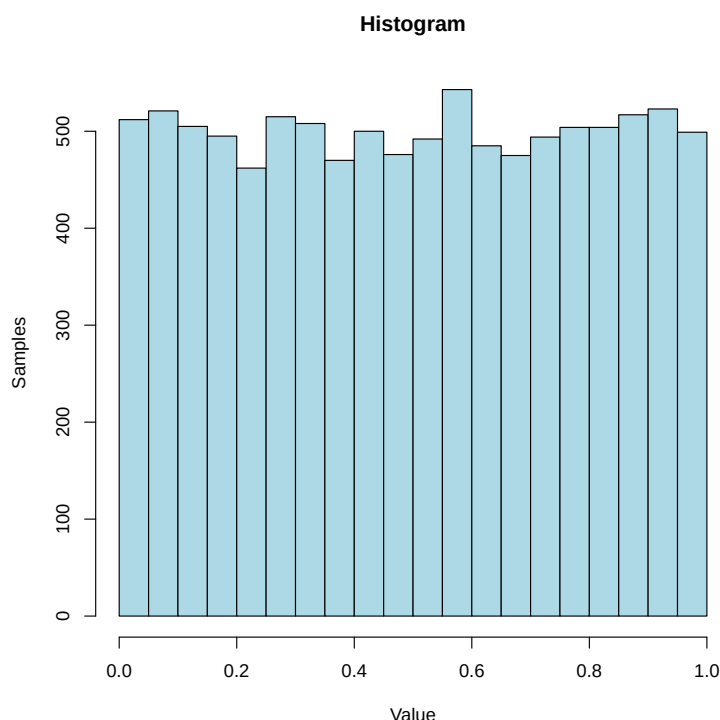


Figure 1: Histogram of 10000 samples generated with `runif()`.

Conclusion: As shown in Figure 1, the samples are approximately evenly distributed across all bins in $[0, 1]$, with no systematic bias toward any sub-interval. This is consistent with the expected behavior of a uniform distribution, confirming that `runif()` is generating numbers as intended.

2.2 Exercise 2

Objective: Verify, using `runif()`, that the sequence generated by a pseudo-random number generator is in fact deterministic, i.e., that it can be reproduced exactly.

Approach: The random-number generator was initialized with seed 123 before generating each of two five-value samples. A third sample was generated without resetting the seed and used as a control.

Conclusion: The two samples generated after resetting the seed are identical, while the control sample differs. Thus, R's pseudo-random sequence is reproducible from a given initial state; it is deterministic rather than truly random.

3 B. Generation of random variables and stochastic processes

3.1 Exercise 3

Objective: Develop an R script to generate Bernoulli distributed numbers from a uniform random number generator (`runif()`), and plot a histogram of 1000 samples to informally check that the generator behaves correctly.

Approach: A Bernoulli outcome with success probability $p = 0.7$ was obtained by thresholding a uniform random value at p . The procedure was repeated for 1000 samples, and the two outcome frequencies were plotted.

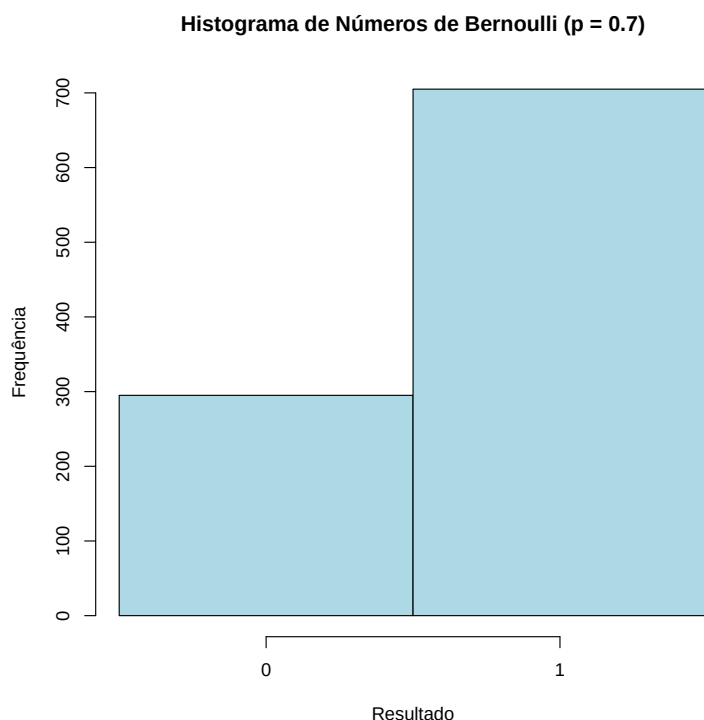


Figure 2: Histogram of 1000 Bernoulli samples with $p = 0.7$.

Conclusion: As shown in Figure 2, out of 1000 samples approximately 700 outcomes are 1 and 300 are 0, matching the expected proportions for $p = 0.7$. This confirms that thresholding a uniform random variable at p is a valid method for generating Bernoulli distributed samples, since $P(U < p) = p$ for $U \sim \text{Unif}(0, 1)$.

3.2 Exercise 4

Objective: Develop an R script to generate exponentially distributed numbers from a uniform random number generator (`runif()`), and plot a histogram of 10000 samples superposed with the theoretical pdf, to informally check that the generator behaves correctly.

Approach: Exponential samples were generated from uniform random values by inverse transform, $X = -\ln(1 - U)/\lambda$. The experiment used $\lambda = 1.5$ and 10000 samples. One histogram shows raw counts; the other is normalized as a density so it can be compared with the theoretical exponential density.

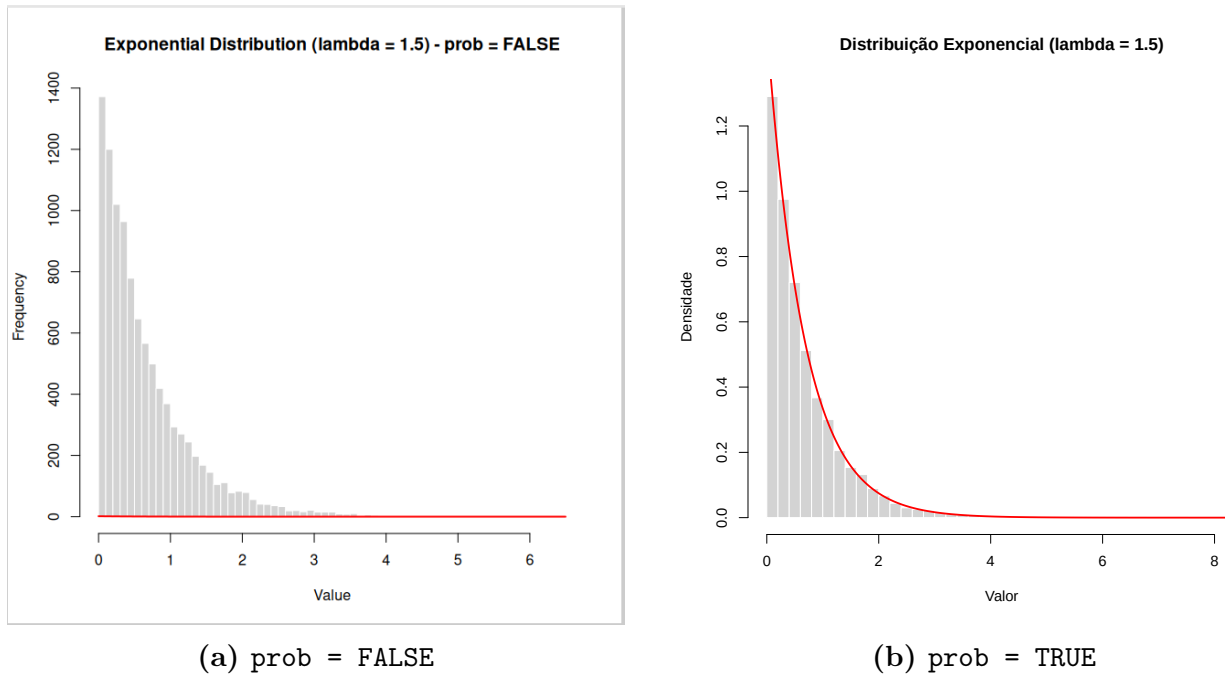


Figure 3: Histogram of 10000 exponentially distributed samples ($\lambda = 1.5$) with the theoretical pdf (red line) superposed, with and without density normalization.

Conclusion: Figure 3 shows why the `prob=TRUE` option is essential for this comparison. In Figure 3a (`prob=FALSE`), the histogram bars represent raw counts per bin, whose total sums to $n = 10000$, while the `dexp()` curve has area 1 by definition; the two are on completely different scales, so the pdf line appears flat against the axis and gives no useful comparison. In Figure 3b (`prob=TRUE`), the histogram is rescaled so that the total area of the bars is also 1, matching the normalization of the theoretical pdf, and the empirical histogram now closely follows the red curve. This confirms that the inverse transform method correctly generates exponentially distributed samples from uniform random numbers, and that density normalization is a necessary condition for a visual histogram-vs-pdf comparison to be meaningful.

3.3 Exercise 5

extbfObjective: Generate the arrival times of a Poisson process and check whether the estimated rate agrees with the configured population rate.

extbfApproach: For a Poisson process with rate λ , inter-arrival times are exponentially distributed with rate λ . Ten thousand inter-arrival times were generated for $\lambda = 2$; cumulative sums give the arrival times, and the reciprocal sample mean estimates the rate.

extbfConclusion: In the recorded run, the estimated rate was approximately 1.992, close to the configured value 2. The reciprocal relation follows because the mean inter-arrival time is $1/\lambda$; agreement is expected to improve with a larger sample, although an individual estimate varies randomly.

4 C. Programming discrete event simulators

4.1 Exercise 6

Objective: Modify the M/M/1 simulator of Annex A to also estimate the average delay in system and the server utilization, and compare the results against the theoretical values (Annex B), for at least two values of λ/μ with $\lambda/\mu < 1$.

Approach: Starting from the simulator in Annex A, the following changes were made to also track the average delay in system and the server utilization, in addition to the average delay in queue already computed by the original code:

- State variables were added to accumulate customers' time in the system and to measure the intervals during which the server is busy.
- At the very start of each loop iteration, right after **Time** is updated and *before* branching on the event type, the interval since the last event is added to **BusyTime** if the server was busy during that interval. This must be done unconditionally for both arrivals and departures, since the server can remain busy across either type of event:

```
NextEventType=which.min(EventList)
Time=EventList[NextEventType]
if (ServerStatus==1) { # NEW
  BusyTime=BusyTime+(Time-LastEventTime) # NEW
} # NEW
LastEventTime=Time # NEW
```

- In the arrival branch, when the server is free, the customer's arrival time is stored in **InServiceArrivalTime** (as before):

```
ServerStatus=1
InServiceArrivalTime=Time # NEW
EventList[2]=Time+rexp(1,ServiceRate)
```

- On departure, the customer's time in the system is accumulated. If another customer is waiting, its arrival time is stored when service begins:

```

AcumSystemTime=AcumSystemTime+
    Time-InServiceArrivalTime
NumQueueCompleted=NumQueueCompleted+1
if (NumInQueue==0) {
    ServerStatus=0
    EventList[2]=Inf
} else {
    AcumDelay=AcumDelay+Time-QueueArrivalTime[1]
    InServiceArrivalTime=QueueArrivalTime[1]
    QueueArrivalTime=QueueArrivalTime[-1]
    NumInQueue=NumInQueue-1
    EventList[2]=Time+rexp(1,ServiceRate)
}

```

- After the loop, the average delay in system and the server utilization are computed alongside the average delay in queue:

```

AvgDelay=AcumDelay/NumQueueCompleted
AvgTimeInSystem=AcumSystemTime/NumQueueCompleted # NEW
Utilization=BusyTime/Time # NEW

```

The simulation was run with $\text{NumQueueCompleted} = 1000$ for three stable configurations ($\lambda/\mu < 1$) and, for comparison, one critically loaded configuration ($\lambda/\mu = 1$). The values in the table are single simulation realizations; because the script does not set a random seed, subsequent runs can produce different estimates. The theoretical values were computed from Annex B: $W_Q = \frac{\lambda}{\mu(\mu - \lambda)}$, $W = \frac{1}{\mu - \lambda}$, and $\rho = \lambda/\mu$.

λ	μ	W_Q (sim.)	W_Q (theo.)	W (sim.)	W (theo.)	ρ (sim.)	ρ (theo.)
1	2	0.4572	0.5000	0.9473	1.0000	0.5421	0.5000
1	3	0.1678	0.1667	0.5142	0.5000	0.2975	0.3333
1	4	0.0727	0.0833	0.3186	0.3333	0.2732	0.2500
2	2	3.3556	∞	3.8212	∞	0.9939	1.0000

Table 1: Simulated vs. theoretical average delay in queue (W_Q), average delay in system (W), and server utilization (ρ), for different λ/μ ratios.

Conclusion: As shown in Table 1, for all stable configurations ($\rho < 1$) the simulated W_Q , W , and ρ closely match the theoretical values predicted by the M/M/1 formulas, with the small discrepancies attributable to the finite sample size (1000 completed customers) and the inherent variability of the simulation. Both the delays and the utilization increase as λ/μ approaches 1: for the M/M/1 system, $\rho = \lambda/\mu$ is precisely the theoretical fraction of time the server is busy, and this matches well against $\text{BusyTime}/\text{Time}$ measured directly from the simulated trajectory. The last row ($\lambda = \mu = 2$, $\rho = 1$) illustrates the boundary of stability: the theoretical delays are infinite in the steady state, since the system is critically loaded (arrivals occur, on average,

as fast as they can be served), yet the simulation returns large but finite delay values together with a simulated utilization close to 1 (0.9939). This finite delay is not a valid estimate of a steady-state quantity; it simply reflects that, over a finite run of 1000 customers, the queue built up substantially – with the server almost always busy – without the delay yet diverging. This confirms that the condition $\rho < 1$ required by the exercise is essential for the M/M/1 system to reach a stable steady state, and shows that server utilization, unlike the mean queuing delay, remains a well-defined and finite quantity even at the stability boundary.

4.2 Exercise 7

Objective: Develop a simulator for estimating the average delay in queue of an M/M/2 system, and compare the results with the theoretical values (Annex C), for at least two values of $\lambda/(2\mu)$ with $\lambda/(2\mu) < 1$.

Approach: The M/M/1 simulator of Annex A was extended to two parallel servers. The key changes are: `ServerStatus` becomes a length-2 vector (one entry per server), and the event list has three entries instead of two – one for arrivals and one departure event per server. On arrival, if both servers are busy the customer joins the queue; otherwise the customer is assigned to the first free server found with `which(ServerStatus==0)[1]`. On a departure, the freed server either goes idle (if the queue is empty) or immediately picks up the next customer in queue, and the corresponding delay is accumulated exactly as in the M/M/1 case.

```

ArrivalRate=1
ServiceRate=2
Time=0
NumQueueCompleted=0
ServerStatus=c(0,0) # 0 = idle, 1 = busy, one entry per server
NumInQueue=0
AcumDelay=0
QueueArrivalTime=c()
EventList=c(rexp(1,ArrivalRate), Inf, Inf) # [arrival, dep@server1, dep@server2]

while (NumQueueCompleted<1000) {
  NextEventType=which.min(EventList)
  Time=EventList[NextEventType]

  if (NextEventType==1) { # ARRIVAL
    EventList[1]=Time+rexp(1,ArrivalRate)
    if (all(ServerStatus==1)) { # both servers busy -> join queue
      QueueArrivalTime=c(QueueArrivalTime,Time)
      NumInQueue=NumInQueue+1
    } else { # a server is free -> start service
      freeServer=which(ServerStatus==0)[1]
      ServerStatus[freeServer]=1
      NumQueueCompleted=NumQueueCompleted+1
      EventList[freeServer+1]=Time+rexp(1,ServiceRate)
    }
  } else { # DEPARTURE from server (NextEventType-1)
    server=NextEventType-1
    if (NumInQueue==0) {
      ServerStatus[server]=0
      EventList[NextEventType]=Inf
    } else {
      AcumDelay=AcumDelay+Time-QueueArrivalTime[1]
      QueueArrivalTime=QueueArrivalTime[-1]
      NumInQueue=NumInQueue-1
      NumQueueCompleted=NumQueueCompleted+1
      EventList[NextEventType]=Time+rexp(1,ServiceRate)
    }
  }
}

AvgDelay=AcumDelay/NumQueueCompleted

```

The simulation was run with $\text{NumQueueCompleted} = 1000$ for two configurations satisfying $\rho = \lambda/(2\mu) < 1$. The values in the table are single simulation realizations; because the script does not set a random seed, subsequent runs can produce different estimates. The theoretical average delay in queue was computed from Annex C: $W_Q = P_Q \frac{\rho}{\lambda(1-\rho)}$, where P_Q is the

probability that an arrival finds both servers busy (Erlang C formula, $m = 2$).

λ	μ	$\rho = \lambda/(2\mu)$	W_Q (sim.)	W_Q (theo.)
1	2	0.25	0.0556	0.0333
2	2	0.50	0.1380	0.1667

Table 2: Simulated vs. theoretical average delay in queue (W_Q) for the M/M/2 system, for different $\lambda/(2\mu)$ ratios.

Conclusion: As shown in Table 2, the simulated W_Q is reasonably close to the theoretical value in both configurations, with the discrepancy attributable to the finite sample size (1000 completed customers) and the inherent variability of a single simulation run – this is discussed further in Exercise 12, which shows that a single run is generally not enough to draw firm conclusions. As with the M/M/1 system, the average delay in queue increases with $\rho = \lambda/(2\mu)$, since higher load makes it more likely that an arriving customer finds both servers busy. Comparing with Exercise 6, for a similar load level the M/M/2 system exhibits substantially lower delays than the corresponding M/M/1 system (e.g., $\rho = 0.5$ gives $W_Q \approx 0.17$ here, versus $W_Q \approx 0.50$ for M/M/1 with $\lambda = 1, \mu = 2$), illustrating the benefit of server pooling: two servers each handling half the traffic intensity of an equivalent single fast server yield lower expected waiting times, since the probability of both being simultaneously busy is lower than the probability of a single server being busy.

5 D. Estimation of probabilities

5.1 Exercise 8

Objective: Estimate, through simulation, the probability of obtaining exactly two heads when five biased coins are tossed, where the probability of obtaining a head is $p = 0.7$. The simulated estimate is then compared with the exact probability.

Approach: Each coin toss was generated from a uniform random variable $U \sim \text{Unif}(0, 1)$. A toss was classified as a head when $U < p$ and as a tail otherwise. This experiment was repeated 1000 times, with five tosses in each experiment. The estimated probability is the fraction of experiments that resulted in exactly two heads. The seed was fixed to make the result reproducible.

```

p_head <- 0.7
n_coins <- 5
n_experiments <- 1000

set.seed(8)

random_values <- matrix(
  runif(n_experiments * n_coins),
  nrow = n_experiments,
  ncol = n_coins
)

coin_tosses <- random_values < p_head
number_of_heads <- rowSums(coin_tosses)

simulation_probability <- mean(number_of_heads == 2)

```

Exact probability: Let X denote the number of heads in five tosses. Since the tosses are independent and have the same probability of success, $X \sim \text{Binomial}(5, 0.7)$. Therefore,

$$P(X = 2) = \binom{5}{2} (0.7)^2 (1 - 0.7)^3 = 0.1323.$$

The simulation and exact results are shown in Table 3.

Simulation probability	Exact probability	Absolute error
0.1480	0.1323	0.0157

Table 3: Simulated and exact probability of obtaining exactly two heads.

Conclusion: The estimated probability is 0.1480, while the exact value is 0.1323. The absolute error is 0.0157, which is expected because the estimate is based on a finite number of experiments and therefore is subject to random variation. Increasing the number of repetitions would generally make the simulated probability closer to the exact value.

6 E. Revisions of statistics

6.1 Exercise 9

Objective: Analyze and compare the packet-size distributions recorded in the Wireshark captures for video, download, and normal-state network activity.

Approach: Traffic was captured with Wireshark in three scenarios – watching a video on YouTube, downloading a large file, and an idle (“normal state”) system – and each capture was exported to CSV via Wireshark’s Export Packet Dissections option. Packet sizes were read from the Wireshark **Length** field; missing and negative values were discarded, but no upper size threshold was applied. This retains the observed 1514-byte records instead of assuming,

without supporting capture evidence, that they are artifacts of TCP Segmentation Offload. All three histograms use common 50-byte bins and a horizontal scale extending to 1550 bytes.

```
packet_sizes <- as.numeric(capture$Length)
packet_sizes <- packet_sizes[!is.na(packet_sizes) & packet_sizes >= 0]

cat("Packets analyzed:", length(packet_sizes), "\n")
cat("Mean:", round(mean(packet_sizes), 2),
    "bytes | Median:", median(packet_sizes), "bytes\n")

hist(packet_sizes,
     breaks = seq(0, 1550, by = 50),
     xlim = c(0, 1550),
     xlab = "Packet size (bytes)", ylab = "Frequency",
     col = "lightblue", border = "white")
```

The results for all three captures are summarized in Table 4.

Capture	Analyzed	Mean (bytes)	Median (bytes)
Video	37,072	1088.96	1292
Download	1,023,788	1121.25	1304
Normal state	53,106	345.21	330

Table 4: Packet-size statistics for all valid Wireshark `Length` values.

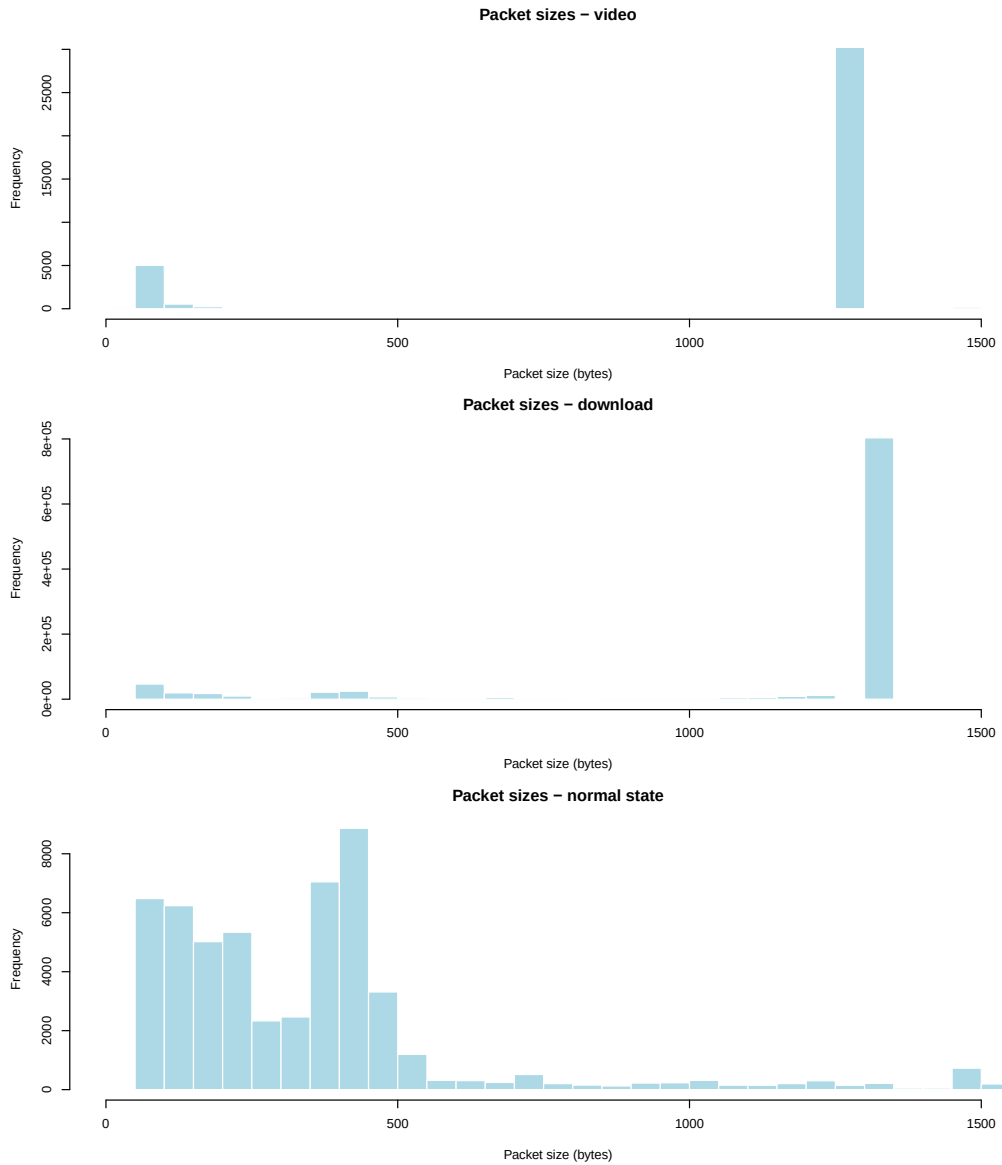


Figure 4: Packet-size histograms for the video, download, and normal-state captures, using common 50-byte bins.

Conclusion: Video and download have similar medians (1292 and 1304 bytes), while normal-state traffic has a substantially smaller median (330 bytes). The means are 1088.96, 1121.25, and 345.21 bytes, respectively. For video and download, the median exceeds the mean, indicating that smaller packet observations pull the mean downward; in the normal-state capture the mean is slightly above the median. The 1514-byte observations are retained as recorded by Wireshark, and their presence alone is not sufficient to identify an offloading artifact. These statistics describe the captured sessions only; they do not establish that the activity itself caused the observed packet-size differences.

6.2 Exercise 10

Objective: Measure 10 Round Trip Times (RTTs) to a distant country and determine two 95% confidence intervals for the average RTT, one based on the Student's t distribution and the other on the Normal distribution, discussing which is preferable.

Approach: The destination, `www.uq.edu.au` (University of Queensland, Australia), was suggested by an LLM-based chatbot that was consulted for a suitable distant host; the measurements themselves were collected by us. Ten RTTs were obtained in a single run of `ping -c 10`, all of them valid replies (Figure 5). The sample mean $\bar{x}$, standard deviation s and standard error $s/\sqrt{n}$ were computed, and the critical points were obtained with `qnorm()` and `qt()`. Both intervals have the form $\bar{x} \pm c \cdot s/\sqrt{n}$, differing only in the critical value c : $z_{0.975}$ for the Normal and $t_{0.975, n-1}$ for Student's t .

```
ataid_jo@JOSELEN2212:~/DDRS/Performance-Evaluation-and-Dimensioning-of-Networks-and-Systems$ ping -c 10 www.uq.edu.au
PING www.uq.edu.au (130.102.184.3) 56(84) bytes of data.
64 bytes from wsa-float-prnpref.lb.dc.uq.edu.au (130.102.184.3): icmp_seq=1 ttl=233 time=325 ms
64 bytes from wsa-float-prnpref.lb.dc.uq.edu.au (130.102.184.3): icmp_seq=2 ttl=233 time=317 ms
64 bytes from wsa-float-prnpref.lb.dc.uq.edu.au (130.102.184.3): icmp_seq=3 ttl=233 time=316 ms
64 bytes from wsa-float-prnpref.lb.dc.uq.edu.au (130.102.184.3): icmp_seq=4 ttl=233 time=316 ms
64 bytes from wsa-float-prnpref.lb.dc.uq.edu.au (130.102.184.3): icmp_seq=5 ttl=233 time=316 ms
64 bytes from wsa-float-prnpref.lb.dc.uq.edu.au (130.102.184.3): icmp_seq=6 ttl=233 time=317 ms
64 bytes from wsa-float-prnpref.lb.dc.uq.edu.au (130.102.184.3): icmp_seq=7 ttl=233 time=316 ms
64 bytes from wsa-float-prnpref.lb.dc.uq.edu.au (130.102.184.3): icmp_seq=8 ttl=233 time=317 ms
64 bytes from wsa-float-prnpref.lb.dc.uq.edu.au (130.102.184.3): icmp_seq=9 ttl=233 time=316 ms
64 bytes from wsa-float-prnpref.lb.dc.uq.edu.au (130.102.184.3): icmp_seq=10 ttl=233 time=321 ms

--- www.uq.edu.au ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9706ms
rtt min/avg/max/mdev = 315.744/317.745/325.308/2.977 ms
ataid_jo@JOSELEN2212:~/DDRS/Performance-Evaluation-and-Dimensioning-of-Networks-and-Systems$ |
```

Figure 5: Output of `ping -c 10 www.uq.edu.au`.

```
rtt <- c(325, 317, 316, 316, 316, 317, 316, 317, 316, 321) # RTT samples (ms)

n <- length(rtt)
xbar <- mean(rtt)
s <- sd(rtt)
se <- s / sqrt(n) # standard error of the mean

alpha <- 0.05 # 95% confidence
z_crit <- qnorm(1 - alpha/2) # Normal critical point
t_crit <- qt(1 - alpha/2, df = n - 1) # Student's t critical point

ci_normal <- c(xbar - z_crit * se, xbar + z_crit * se)
ci_t <- c(xbar - t_crit * se, xbar + t_crit * se)
```

n	$\bar{x}$ (ms)	s (ms)	$s/\sqrt{n}$ (ms)	Critical value	95% CI (ms)
10	317.70	2.98	0.94	$z = 1.960$ (Normal)	[315.85, 319.55]
10	317.70	2.98	0.94	$t_9 = 2.262$ (Student)	[315.57, 319.83]

Table 5: 95% confidence intervals for the average RTT using the Normal and Student's t distributions.

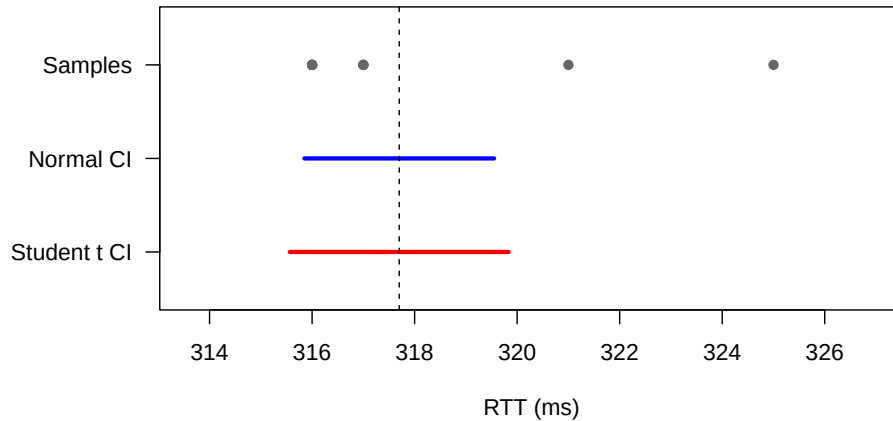


Figure 6: RTT samples and the two 95% confidence intervals (dashed line: sample mean).

Conclusion: As shown in Table 5 and Figure 6, the Student's t interval is wider than the Normal interval (4.27 ms versus 3.70 ms), since $t_9 = 2.262 > z = 1.960$. With a small sample and unknown population standard deviation, the Student's t interval is generally preferable under the assumption that the observations are independent and approximately Normal. The Normal interval here uses the sample standard deviation in place of the unknown population value; its actual coverage depends on the population distribution and cannot be asserted from this sample alone. Eight observed RTTs lie between 316 and 317 ms, while 321 and 325 ms are higher. These ten measurements are not enough to establish that the underlying RTT distribution is non-Normal or right-skewed, so the shape should be treated cautiously; Exercise 11 studies interval coverage under known distributions.

6.3 Exercise 11

Objective: For four distributions – standard Normal ($\mu = 0, \sigma^2 = 1$), Exponential ($\lambda = 1$), Lognormal ($\mu = 0, \sigma^2 = 1$) and Lognormal ($\mu = 0, \sigma^2 = 2$) – and for sample sizes $n = 5, 10, 50, 5000$, estimate the coverage of the 95% confidence interval for the mean based on 1000 experiments, and discuss whether these intervals can be trusted.

Approach: For each distribution and each sample size n , 1000 samples of size n were generated. For each sample, the 95% CI for the mean was computed as $\bar{x} \pm t_{0.975, n-1} \cdot s/\sqrt{n}$, using the Student's t critical point, and the coverage was estimated as the fraction of the 1000 intervals that contained the true population mean μ . For the exponential distribution $\mu = 1/\lambda = 1$, and for the lognormal distributions $\mu = e^{\mu_{\log} + \sigma^2/2}$, giving $\mu \approx 1.649$ for $\sigma^2 = 1$ and $\mu \approx 2.718$ for $\sigma^2 = 2$.

The table and figure report one Monte Carlo realization. Since the script does not set a random seed, the exact coverage estimates may differ when it is run again.

```
E <- 1000 # number of experiments
SizesOfSamples <- c(5, 10, 50, 5000) # sizes of samples
tcritical <- qt(0.975, df = SizesOfSamples - 1) # critical points, t
distribution

# Estimates the CI coverage for one distribution
# rgen: function(N) that generates a sample of size N
# mu: true (population) mean
coverage <- function(rgen, mu) {
  InInt <- rep(0, length(SizesOfSamples))
  for (i in seq_along(SizesOfSamples)) {
    N <- SizesOfSamples[i]
    for (j in 1:E) {
      a <- rgen(N)
      ma <- mean(a)
      va <- var(a)
      alinf <- ma - tcritical[i] * sqrt(va / N)
      alsup <- ma + tcritical[i] * sqrt(va / N)
      if (mu > alinf & mu < alsup) InInt[i] <- InInt[i] + 1
    }
  }
  return(InInt / E)
}

# Standard Normal (mu = 0, sigma^2 = 1): mean = 0
cov_norm <- coverage(function(N) rnorm(N), mu = 0)

# Exponential (lambda = 1): mean = 1/lambda = 1
cov_exp <- coverage(function(N) rexp(N, rate = 1), mu = 1)

# Lognormal (mu = 0, sigma^2 = 1): mean = exp(mu + sigma^2/2)
cov_ln1 <- coverage(function(N) rlnorm(N, meanlog = 0, sdlog = 1), mu = exp(0 +
  1/2))

# Lognormal (mu = 0, sigma^2 = 2)
cov_ln2 <- coverage(function(N) rlnorm(N, meanlog = 0, sdlog = sqrt(2)), mu =
  exp(0 + 2/2))
```

n	Normal	Exponential	Lognormal ($\sigma^2 = 1$)	Lognormal ($\sigma^2 = 2$)
5	0.951	0.898	0.807	0.699
10	0.960	0.899	0.846	0.746
50	0.957	0.930	0.904	0.815
5000	0.950	0.953	0.951	0.940

Table 6: Estimated coverage of the nominal 95% confidence interval for the mean, for four distributions and four sample sizes (1000 experiments each).

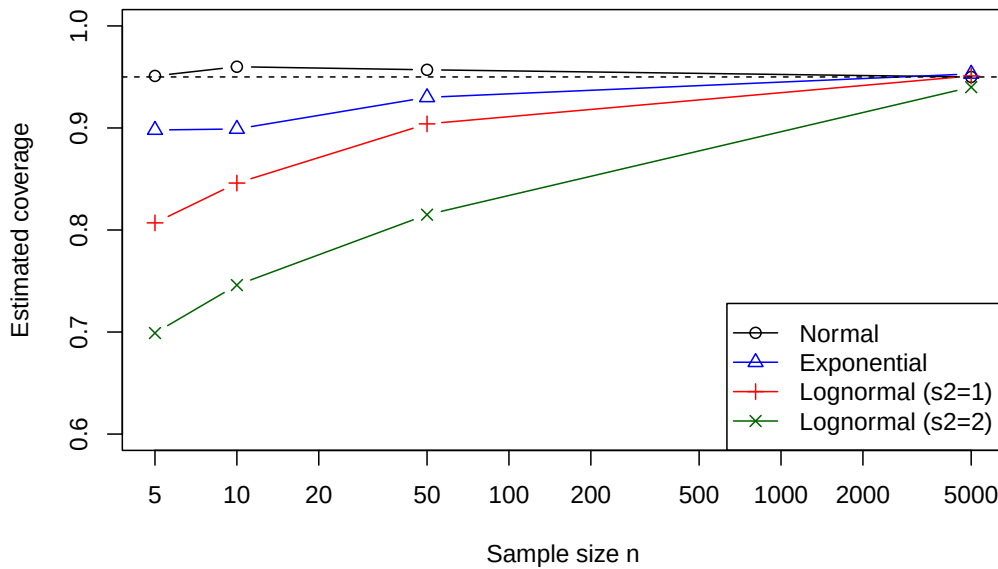


Figure 7: Estimated coverage vs. sample size n (log scale), for the four distributions. Dashed line: nominal 95% level.

Conclusion: The results illustrate that empirical coverage depends on both the sample size and the distribution. For the Normal population, the observed coverages are close to the nominal 95% at all four sample sizes. For the skewed populations they are generally lower at small sample sizes and move closer to 95% as n increases, although the estimate at $n = 5000$ for the lognormal with $\sigma^2 = 2$ is still 0.940. This last difference should be interpreted in light of Monte Carlo uncertainty: with 1000 experiments, an estimated coverage near 0.95 has a standard error of about 0.007. The results are consistent with the Central Limit Theorem improving the Normal approximation for larger samples, but a single set of 1000 experiments does not prove exact coverage or systematic bias.

7 F. Output data analysis

7.1 Exercise 12

Objective: Assess the variability of the M/M/1 average queue-delay estimate by running the simulator ten times, with 1000 completed customers per run and traffic intensity $\rho = \lambda/\mu = 0.8$, and determine whether a single simulation is sufficient to draw conclusions.

Approach: The event-driven logic in `mm1.R` was parameterized in the Exercise 12 script because that helper is a standalone script with fixed input values. Ten independent runs used $\lambda = 0.8$ and $\mu = 1$, giving $\rho = 0.8$. Each estimate is the accumulated queueing delay divided by the number of customers crossing the queue; customers that enter service immediately contribute zero. A fixed random seed (12) makes the experiment reproducible. The simulation loop was repeated as follows:

```
set.seed(12)
queue_delay_estimates <- replicate(
  10,
  simulate_mm1_queue_delay(
    arrival_rate = 0.8,
    service_rate = 1,
    completed_customers = 1000
  )
)
```

The theoretical steady-state average queue delay is

$$W_Q = \frac{\lambda}{\mu(\mu - \lambda)} = \frac{0.8}{1(1 - 0.8)} = 4.$$

The ten simulation estimates are shown in Table 7.

Run	W_Q estimate	Run	W_Q estimate
1	2.7141	6	2.3415
2	4.0146	7	4.9274
3	2.7726	8	3.4723
4	3.7525	9	2.4955
5	3.2918	10	3.9343

Table 7: Average queue-delay estimates from ten independent M/M/1 simulation runs (1000 completed customers per run).

Across the ten replications, the sample mean was 3.3717 and the sample standard deviation was 0.8118. The estimates ranged from 2.3415 to 4.9274. A 95% Student's t confidence interval for the mean across replications was [2.7909, 3.9524], which does not contain the theoretical value $W_Q = 4$; this illustrates the uncertainty remaining with only ten replications.

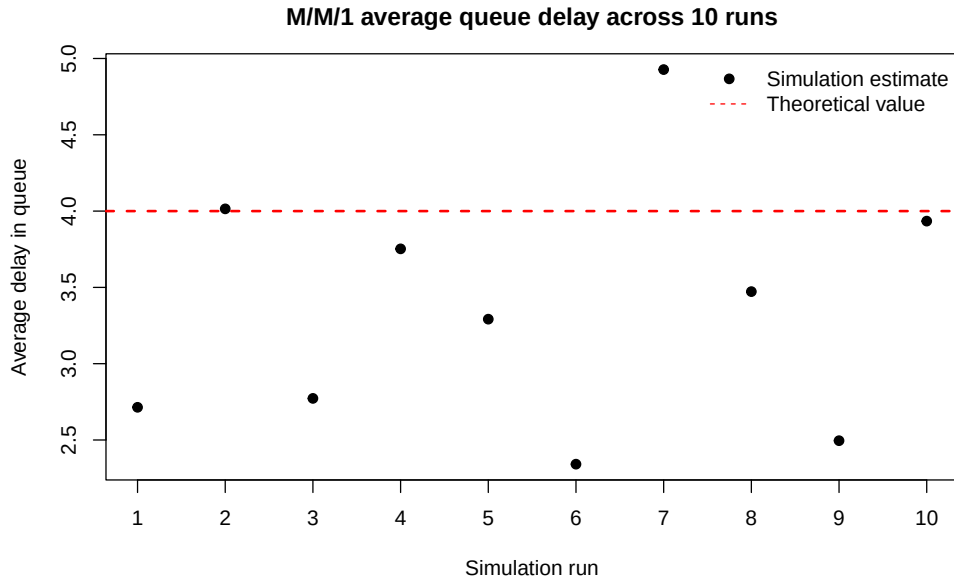


Figure 8: Average queue-delay estimate in each of the ten M/M/1 runs. The dashed line marks the theoretical value $W_Q = 4$.

Conclusion: The estimates differ noticeably from run to run even though each run processes 1000 completed customers. They range from 2.3415 to 4.9274, so a single run cannot characterize the estimator’s variability. The mean across the ten runs is 3.3717, and its 95% confidence interval [2.7909, 3.9524] does not contain the theoretical value 4. This does not contradict the steady-state formula: with only ten replications, the estimate of the mean itself remains uncertain, and this particular experiment produced a low interval. More independent replications or longer runs are needed for a more stable comparison.

7.2 Exercise 13

Objective: Evaluate how the initial number of customers in an M/M/1 system affects estimates of the average delay in queue. Twenty-five independent runs were performed for each combination of run length (20 or 2000 completed customers) and initial population (0 or 10 customers), with $\lambda = 3$ and $\mu = 4$.

Approach: The supplied `fmm1_initcond()` function in `initcond.R` was used for every run. It creates the initial population using successive exponential inter-arrival times, then simulates until the requested number of customers has crossed the queue. The four combinations of run length and initial population were generated with 25 independent replications each:

```

set.seed(13)
run_results <- expand.grid(
  run = 1:25,
  run_length = c(20, 2000),
  initial_clients = c(0, 10)
)
run_results$average_queue_delay <- mapply(
  fmm1_initcond,
  ArrivalRate = 3,
  ServiceRate = 4,
  NumEndClients = run_results$run_length,
  InitClients = run_results$initial_clients
)

```

For $\rho = \lambda/\mu = 0.75$, the theoretical steady-state mean delay in queue is

$$W_Q = \frac{\lambda}{\mu(\mu - \lambda)} = \frac{3}{4(4 - 3)} = 0.75.$$

Table 8 reports the mean of the 25 simulation estimates for each condition and the corresponding 95% Student's t confidence interval across replications.

Completed customers	Initial customers	Mean W_Q	95% confidence interval
20	0	0.6183	[0.3648, 0.8718]
20	10	2.2925	[1.9915, 2.5935]
2000	0	0.6983	[0.6515, 0.7451]
2000	10	0.7692	[0.6984, 0.8399]

Table 8: M/M/1 average queue-delay estimates over 25 runs per condition ($\lambda = 3$, $\mu = 4$).

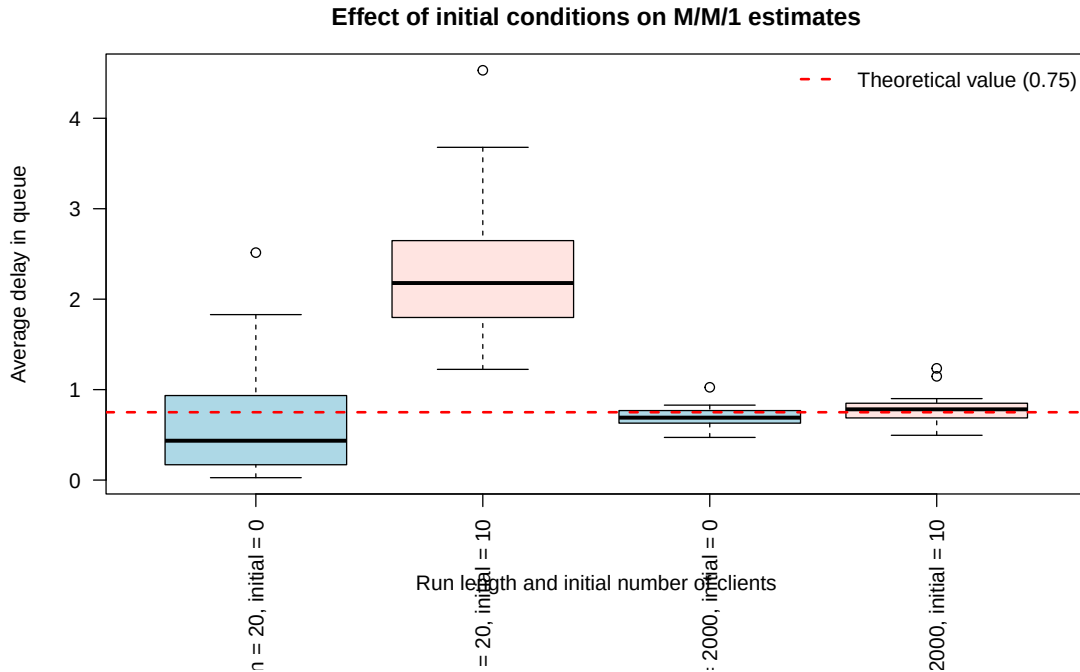


Figure 9: Distribution of the 25 average queue-delay estimates for each run length and initial condition. The dashed line marks the theoretical value $W_Q = 0.75$.

Conclusion: With only 20 customers, the initial population has a pronounced effect: the mean estimate is 0.6183 when starting empty and 2.2925 when starting with ten customers. Under the helper’s initialization, the initial queue is formed through successive arrivals before the main simulation loop, so the initial waiting times strongly affect this short run. With 2000 customers, both estimates (0.6983 and 0.7692) are close to the theoretical value 0.75, and the initial-condition effect is much smaller. The confidence intervals show the same pattern: the short-run interval for ten initial customers is well above 0.75, whereas both long-run intervals include or nearly include the theoretical value (the empty-start interval ends at 0.7451). Longer runs therefore reduce, but do not eliminate in every finite sample, the initial-transient effect.

7.3 Exercise 14

Objective: Study how the number of independent replications required to estimate the M/M/1 average queue delay depends on the target relative error, the number of measured delays per replication, and the traffic intensity $\rho = \lambda/\mu$.

Approach: The supplied `frepdel()` function in `repdel.R` was used. It starts with ten independent replications, deletes the first 100 queue delays in each replication, and adds replications until the relative half-width of the 95% Student’s t confidence interval satisfies the helper’s criterion:

$$\frac{t_{0.975, R-1} s_R}{\sqrt{R} |\bar{W}_{Q,R}|} \leq \frac{\gamma}{1 + \gamma},$$

where R is the number of replications, s_R is the sample standard deviation of the replication estimates, and γ is the requested relative error. We used $\mu = 1$, $\rho \in \{0.50, 0.80, 0.95\}$, measured

sample sizes $n \in \{100, 1000\}$, and $\gamma \in \{5\%, 10\%, 20\%\}$. The theoretical mean and variance of one steady-state M/M/1 queue delay are

$$E[W_Q] = \frac{\rho}{\mu - \lambda}, \quad \text{Var}(W_Q) = \frac{\rho(2 - \rho)}{(\mu - \lambda)^2}.$$

For $\mu = 1$, the theoretical variances are 3, 24, and 399 for $\rho = 0.50, 0.80$, and 0.95 , respectively. Figure 10 shows both this increase and the observed replication requirements.

ρ	Delays per replication	$\gamma = 5\%$	$\gamma = 10\%$	$\gamma = 20\%$
0.50	100	550	103	36
0.80	100	1102	356	116
0.95	100	1036	314	73
0.50	1000	37	12	10
0.80	1000	243	76	19
0.95	1000	771	227	53

Table 9: Number of replications required to reach each relative-error target using a 95% confidence interval. At least ten replications were used in every case.

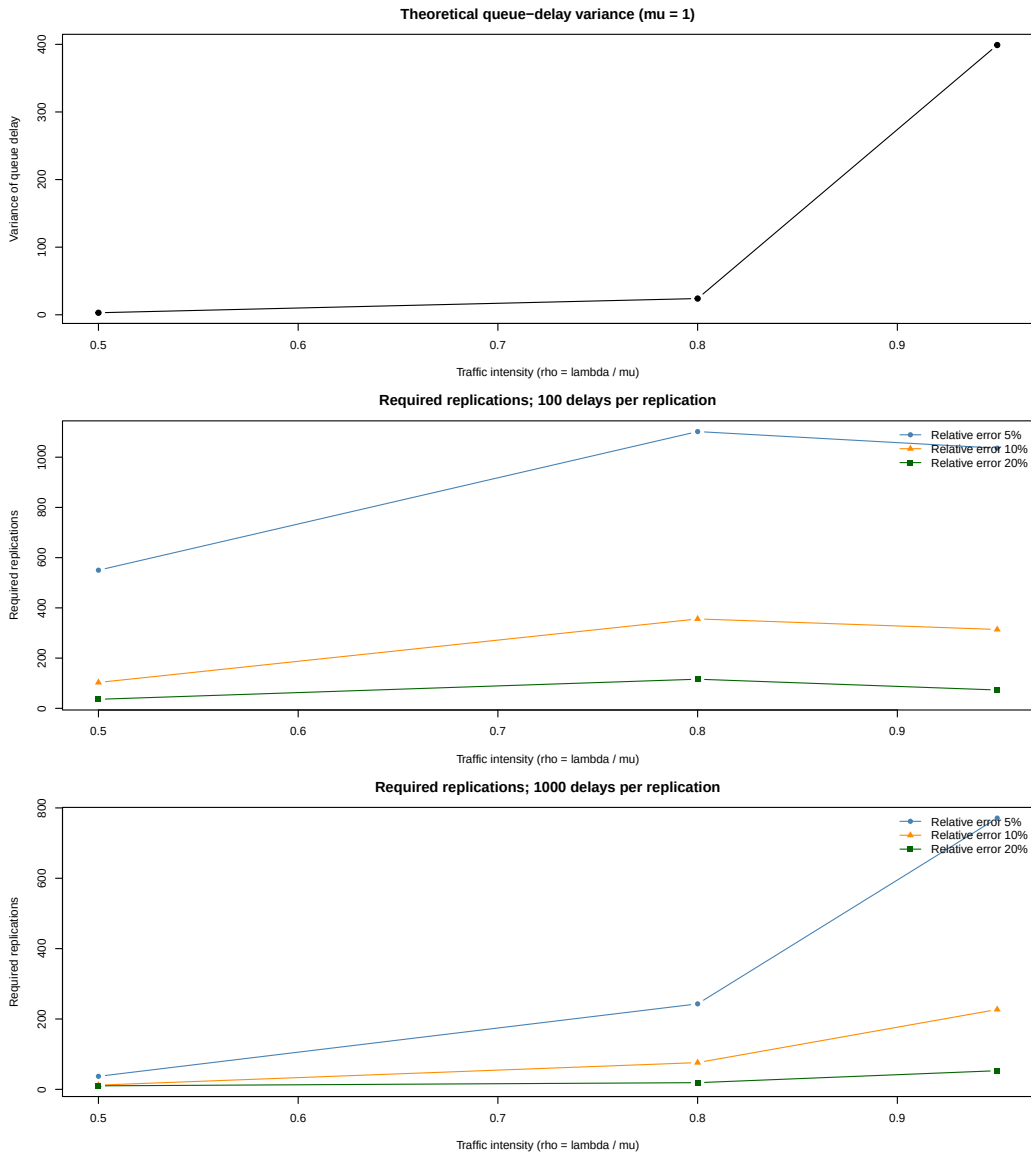


Figure 10: Theoretical queue-delay variance and observed replication requirements as functions of ρ , for two measured sample sizes.

Conclusion: The table shows that stricter relative-error targets generally require more replications, while measuring more delays per replication generally reduces the count. Queue-delay variance increases sharply with load, from 3 at $\rho = 0.50$ to 399 at $\rho = 0.95$, so high-load estimates are intrinsically more variable. The measured counts are stochastic and are not monotonic in ρ for every row; therefore, this experiment supports the role of variability and precision in replication effort but does not establish that the required count rises monotonically with traffic intensity in every finite run.

8 Conclusion

This laboratory work covered the full pipeline needed to build and validate a discrete-event simulator: generating random numbers and random variables, using them to drive event-driven simulation models, and applying statistical techniques to interpret the resulting output correctly.

Exercises 1–2 confirmed that R’s random number generator is uniformly distributed yet fully deterministic given a seed, a property essential to the reproducibility of every simulation in this report. Exercises 3–5 showed that Bernoulli, exponential, and Poisson-process samples can all be derived from this single uniform generator, with their empirical distributions matching the corresponding theoretical ones.

Exercises 6–7 used these techniques to build M/M/1 and M/M/2 simulators. For stable configurations ($\rho < 1$), simulated delays and utilization closely matched queuing theory, while the critically loaded case ($\rho = 1$) showed that a finite simulated delay at the stability boundary does not estimate a (theoretically infinite) steady-state quantity. Comparing the two systems also illustrated the benefit of server pooling, with two parallel servers achieving lower delay than a single server of equivalent aggregate capacity.

Exercise 8 demonstrated Monte Carlo estimation of a probability, converging toward the exact binomial value as expected. Exercise 9 applied similar tools to real captured traffic, showing that bulk transfers (video, downloads) produce large, near-MTU packets, while idle traffic is dominated by small control messages.

Exercises 10–11 addressed the reliability of confidence intervals: the RTT measurements favored the more conservative Student’s t interval for small samples, and the coverage study quantified why, showing that intervals built from skewed distributions with small n cover the true mean far less often than the nominal 95% suggests, only becoming trustworthy once n is large enough for the CLT approximation to hold.

Finally, Exercises 12–14 examined practical challenges of output analysis. Exercise 12 illustrated run-to-run variability in finite simulations. Exercise 13 showed that initial conditions can strongly affect short-run estimates, with a smaller effect in longer runs. Exercise 14 showed that replication effort depends on the precision target, the number of measured delays per replication, and the traffic intensity; the increasing theoretical queue-delay variance near saturation motivates additional replication, although the finite-run counts need not increase monotonically for every tested case.

Taken together, the exercises show that a correct simulation result depends not only on a correctly implemented model, but also on a sound statistical treatment of its output – adequate replication, awareness of initial-transient effects, and confidence intervals appropriate to the shape of the underlying data.